VGC META-GATEKEEPER

Predicting the competitive meta



Can Pokémon attributes identify the species most likely to shape the VGC World Championships?

THE QUESTION

Which Pokémon cross the meta gate?

A leakage-safe, interpretable baseline for rare competitive picks.

Team number : 30

Entry number	Name	Role
2026MCB1541	MALLA DHARANI CHAND	Execution Lead
2026AIB1047	Kanishk	PPT Lead
2026DAB1297	N SHASANK KUMAR REDDY	Documentation Lead
2026EEB1387	PALAGIRI BHAVESH	Video Lead

Git Hub repo link:
https://github.com/2026dab1297-dotcom/vgc-meta-predictor.git

Define the competitive gate

Official VGC doubles

Players bring a team and select four; two Pokémon battle at a time.

1,000+ species

Only a small adapted core appears consistently at the highest level.

RESEARCH QUESTION

Can static attributes predict Worlds meta usage?

Target: META = 1 when Worlds usage is at least 1%.

Rare-positive warning

Accuracy can hide a model that predicts “not meta” for everyone. Recall and F1 matter more.

WHY IT MATTERS

Prepare faster

Focus practice on likely tournament threats.



Balance smarter

Track fairness and competitive diversity.



Build a baseline

Reuse the method in future formats and games.

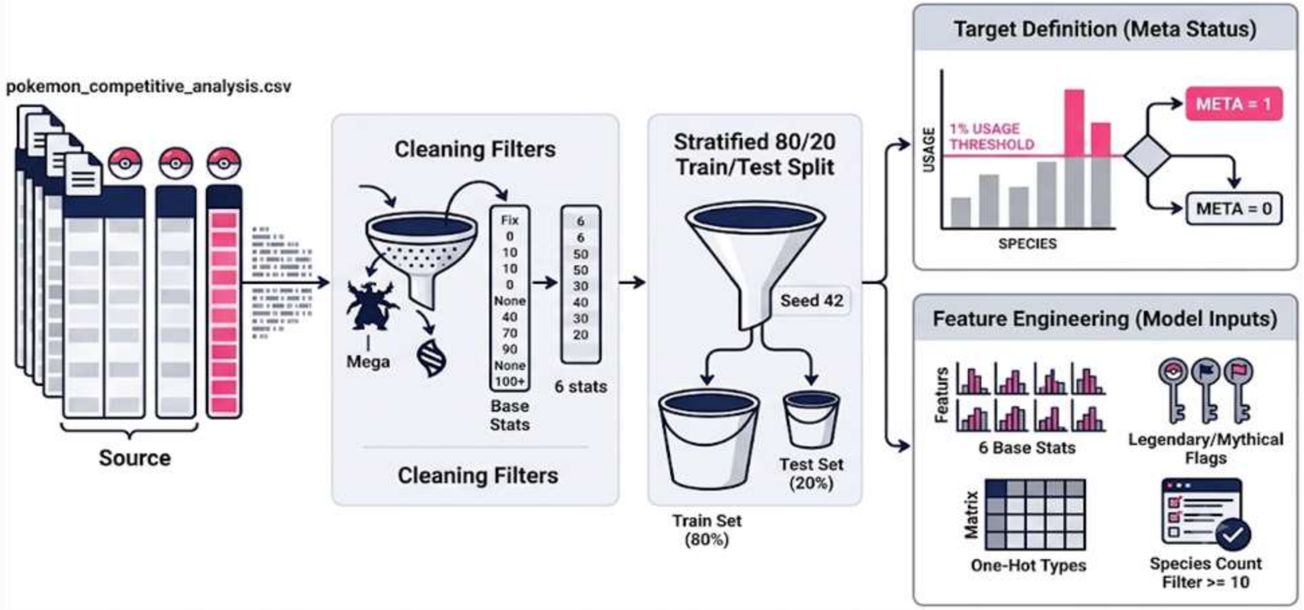


Dataset Information :

Dataset Source:
https://www.kaggle.com/datasets/danielsmdev/pokemon-competitive-usage-smogon-and-vcgworlds

Dataset Size:
Original : 1303 rows x 23 columns
After removing Mega and Gigantamax forms: 1221 x 23

Build a leakage-safe baseline



Libraries used:

- pandas
- NumPy
- scikit-learn: StandardScaler, SelectKBest ,Logistic Regression, cross-validation
- Matplotlib
- Seaborn for analysis plots

LOGISTIC REGRESSION

Features × weights → sigmoid → P(meta)

Positive coefficients raise predicted probability; negative coefficients lower it. Fast, interpretable baseline.

PIPELINE

1 Scale features
2 Select top 25 + six base stats
3 Weight rare-class errors
4 Tune C with 5-fold F1 cross-validation

EDA / coefficient signal: offense and tempo may matter; Special Attack raises probability, while legendary status lowers it in this dataset. Association is not causation.

What the model catches—and misses



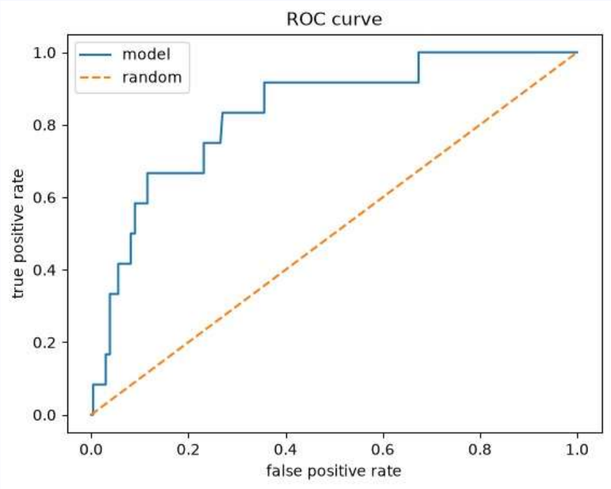



TEST SET CONFUSION MATRIX

Actual class	Predicted not meta	Predicted meta
Not meta	**192**	**41**
Meta	**4**	**8**

From these counts: accuracy 81.6% · precision 16.3% · recall 66.7% · F1 26.2%

Minority-class errors remain important: 4 missed metas and 41 false alarms.



	train	test
accuracy	0.817	0.816
precision	0.192	0.163
recall	0.872	0.667
f1	0.314	0.262
rmse	0.354	0.356
auc	0.919	0.835

baseline accuracy (always say not meta): 0.951

CASE STUDIES

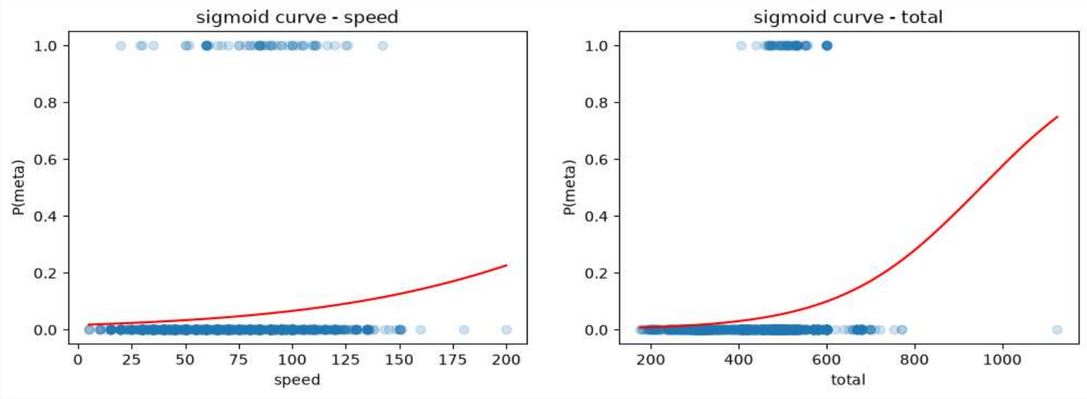
HITS

Incineroar — Intimidate is a visible, model-friendly signal.
Tyranitar — high base stats crossed the learned gate.

MISSES

Rillaboom — 29.72% usage; predicted P(meta) ≈ 0.27.
Corviknight — 2.43% usage; role context was missed.

The model sees attributes, but not moves, team roles, or matchup context.



Turn the baseline into better preparation

WHAT IT CAN SUPPORT

Use likely threats to focus team-building, matchup drills, and practice time. Track whether competitive variety is expanding or shrinking.

What Worked :

Recall of 66.7% and AUC of 0.835 show the model catches most meta Pokémon from stats alone.

What didn't work:

Precision of only 16.3%, with 41 false alarms. Stats alone can't capture moves or team roles.

MODEL LIMITS

Cross-validation chose C = 10 (5-fold F1 ≈ 0.25), meaning weak regularization worked best. Predictions used the default 0.5 cutoff, which may not be the best choice. The threshold should reflect the cost of missed metas versus false alarms.

WHAT TO ADD NEXT

Moves and item data
Team-role signals such as Fake Out, Grassy Glide, and Tailwind
Multiple Worlds seasons
Random Forest and boosting comparisons

WHO DID WHAT

- Malla Dharani Chand - Execution Lead
- Kanishk - PPT Lead
- N Shasank Kumar Reddy - Documentation Lead
- Palagiri Bhavesh - Video Lead

A repeatable baseline is a starting point. Better context and more seasons can help the model learn rare competitive picks.

AI – USE DISCLOSURE:

AI TOOLS USED :

Gemini: used for clearing doubts and concepts while learning.
Claude (Anthropic): used for code clarification (understanding what the code does) and as a reference for drafting the README file, which the team then edited.
ChatGPT (OpenAI): used to cross-check our work.
We checked AI output by running the notebook ourselves, comparing results with what we expected, and reviewing the code as a team. All code is understood by the team and explained in the video.
No AI was used to generate the dataset or the results; all numbers come from running the notebook on the Kaggle dataset.